GEK-65149

DRIVE SYSTEM INSTRUCTIONS

Regulator Test Module

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REV. 1 12/15/78	REV. 2	APPROVALS	GENERAL ELECTRIC	REGULATOR TEST MODULE	
ISSUED February 1, 1978	PRINTS TO	E. Y. Linn	SALEM, VA. U.S.A.	277A4909 CONT. ON SH. 2	SH. NO 1
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REGULATOR TEST MODULE (DS3820RTM).

1. PURPOSE OF EQUIPMENT. The Regulator Test Module (RTM) provides a means of performing basic troubleshooting of a regulator. The module offers three test references and a test mode control. Two of the test references (REF 1 and REF 2) are adjustable, general purpose reference outputs. The other test reference (SUPP) provides a high current, short-duration pulse for firing circuit suppression level testing. All adjustable test reference outputs require jumpering between module test jacks and card test points in the drive under test.

   The TEST mode provides a go/no go test of the regulator amplifiers. Test mode inputs unsuicide the drive, open the motor contactor and convert the regulator loop integrators to fixed gain amplifiers. While in the test mode, the drive is tested without motor feedback signals. Power to this module comes from the regulator electronics module into which it is inserted.

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2. PHYSICAL DESCRIPTION. The Regulator Test Module (RTM) (See Figures 1 and 2) consists of a 6-sided enclosure which contains the module's components. Two potentiometers, two switches and four test jacks are mounted on the RTM front panel. A black lock-in lever located at the bottom of the front panel is linked to the test receptacle and electrically connects and mechanically secures the RTM to the test receptacle. The RTM is supplied with a plastic storage bag.

A. Dimensions:

Length: 8 inches (12.3 cm)
Height: 3 7/8 inches (9.9 cm)
Width: 1 3/4 inches (4.4 cm)
Weight: 1.3 pounds (0.6 kg)

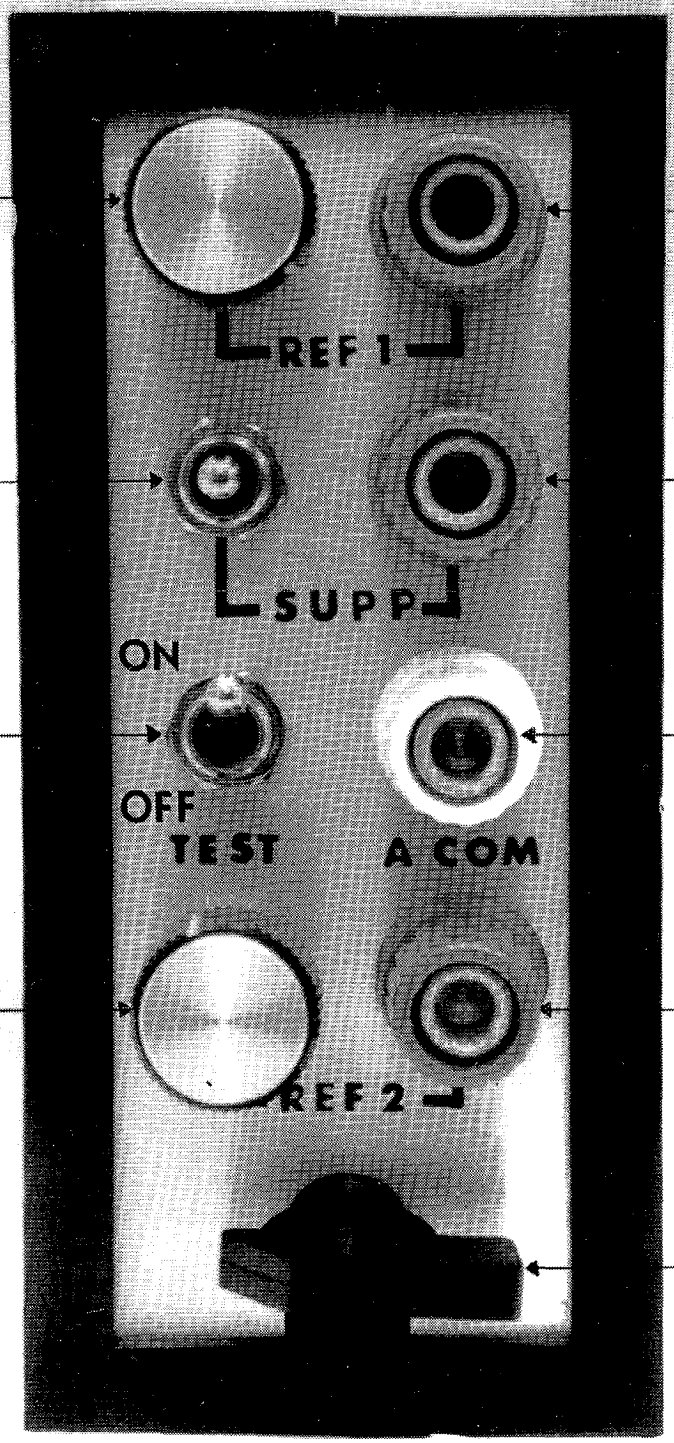
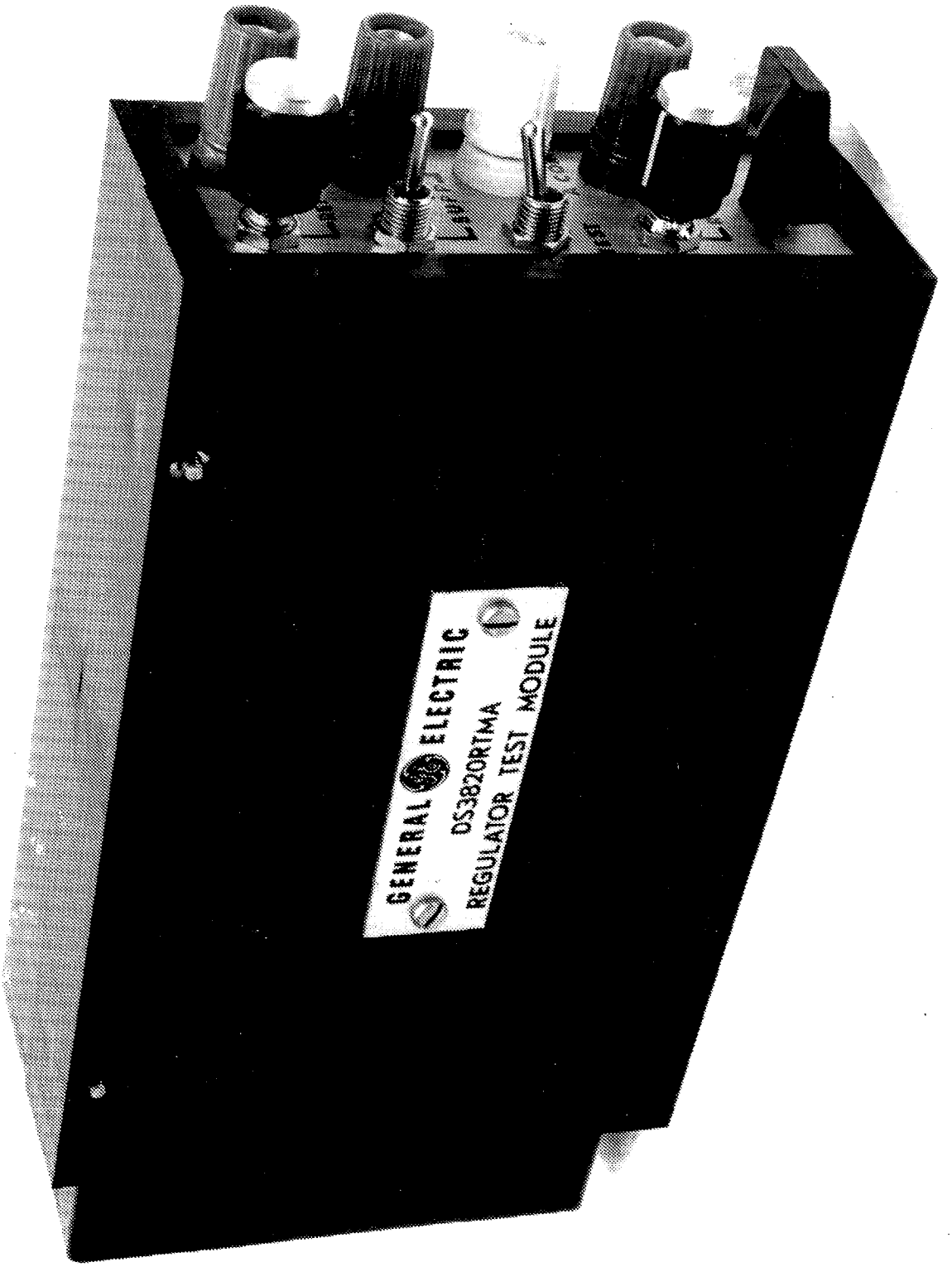


Figure 1 Regulator Test Module (DS3820RTMA) Front Panel

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(ICF49860)

Figure 2 Regulator Test Module (Side View)

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3. ADDITIONAL EQUIPMENT REQUIRED. Test leads (a minimum of two) are required to jumper between RTM test jacks and card test points in the drive electronics. Three test leads are required if two test reference output voltages are applied simultaneously. A voltmeter capable of measuring up to ±15 volts DC is required. If detailed operations such as system tuneup are to be performed, more sophisticated equipment such as a Drive Test Module (DTM), oscilloscope, chart recorder, etc. are recommended.

4. CONNECTING REGULATOR TEST MODULE TO TEST RECEPTACLE.

CAUTION

IMPROPER USE OF THE RTM CAN RESULT IN DAMAGE TO THE SYSTEM. USE OF TEST REFERENCES (REF 1 AND REF 2) ARE INTENDED FOR SIGNAL INJECTION TO OP-AMP TEST INPUTS. INJECTION OF SIGNALS TO DIGITAL TEST POINTS OR TO AN OP-AMP SUMMING JUNCTION IS NOT RECOMMENDED. THE TEST REFERENCE OUTPUTS ARE PROTECTED FROM SHORTS TO GROUND. POTENTIAL DIFFERENCES OF UP TO 43 VOLTS CAN BE WITHSTOOD WITHOUT DAMAGE TO THE MODULE. SWITCHING ON THE TEST MODE SWITCH WITH THE DRIVE MOVING WILL LIKELY RESULT IN DAMAGE TO THE SYSTEM DUE TO THE LOSS OF DRIVE REGULATION. THEREFORE, SWITCHING ON THE TEST MODE SWITCH SHOULD BE DONE WITH THE DRIVE STOPPED. THE SUPPRESSION TEST OUTPUT IS DESIGNED FOR SIGNAL INJECTION TO THE HIGH VOLTAGE BOARD BURDEN RESISTOR. ANY OTHER USE OF THIS OUTPUT IS NOT RECOMMENDED. BEFORE ATTACHING THE RTM TO A DRIVE, THE FOLLOWING STEPS MUST BE PERFORMED:

a. STOP THE DRIVE BY SUICIDING.

b. PLACE THE RTM TEST MODE SWITCH IN THE OFF POSITION.

c. DISCONNECT ALL JUMPER LEADS THAT MAY BE CONNECTED TO THE RTM TEST JACKS.

IF PRECAUTIONS DESCRIBED IN STEPS a THROUGH c ARE NOT TAKEN, INADVERTENT MOVEMENT OR MISOPERATION OF THE DRIVE CAN OCCUR.

To connect the RTM to a test receptacle, refer to Figure 3 and proceed as follows:

(1) Rotate the RTM black actuator lever fully counter-clockwise.

CAUTION

IN THE FOLLOWING STEP, ENSURE PROPER ALIGNMENT OF RTM WITH TEST RECEPTACLE TO AVOID BENDING CONNECTOR PINS.

(2) Carefully engage rear end of RTM over test receptacle.

(3) Firmly hold RTM against test,receptacle and rotate actuator lever fully clockwise (1/4-turn) to vertical position.

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5. RTM CHECKOUT. The reference voltage outputs, reference adjustment potentiometers and grounds are checked out by use of the following tests:

CAUTION

THE RTM CHECKOUT IS ACCOMPLISHED WITHOUT ACCESS TO THE CONNECTOR PINS INSIDE OF THE RTM. THIS IS NECESSARY TO AVOID BENDING THE PINS AND TO PREVENT CONTAMINATION OF THE RTM BY FOREIGN PARTICLES.

In order to perform the following tests, the RTM must be connected to a card rack in which power is applied. To perform the RTM checkout, refer to Figure 4 and proceed as follows:

a. Connect a DC voltmeter to the RTM with the positive lead to the REF 1 jack and the negative lead to the ACOM jack.

b. Rotate REF 1 potentiometer fully clockwise; the meter should indicate +10.74 $\pm$ 1.0 VDC.

c. Rotate REF 1 potentiometer fully counter-clockwise; the meter should indicate -10.74 $\pm$ 1.0 VDC.

d. Adjust REF 1 potentiometer for a 0.00 volt output.

e. Disconnect the positive lead of the voltmeter from the RTM REF 1 test jack and connect it to the REF 2 test jack.

f. Rotate REF 2 potentiometer fully clockwise; the meter should indicate +10.74 $\pm$ 1.0 VDC.

g. Disconnect the positive lead of the voltmeter from the REF 2 jack and connect it to the SUPP test jack.

NOTE

The suppression circuit can be connected to the firing circuit either externally by jumper, or internally through the card rack backplane. The latter connection is a requisition option. In the following step, if the suppression circuit is internally connected through the firing circuit card, depressing the SUPP switch will cause the suppression circuit to be tripped. When tripped, a suppression indicator (LED) in the firing circuit extinguishes.

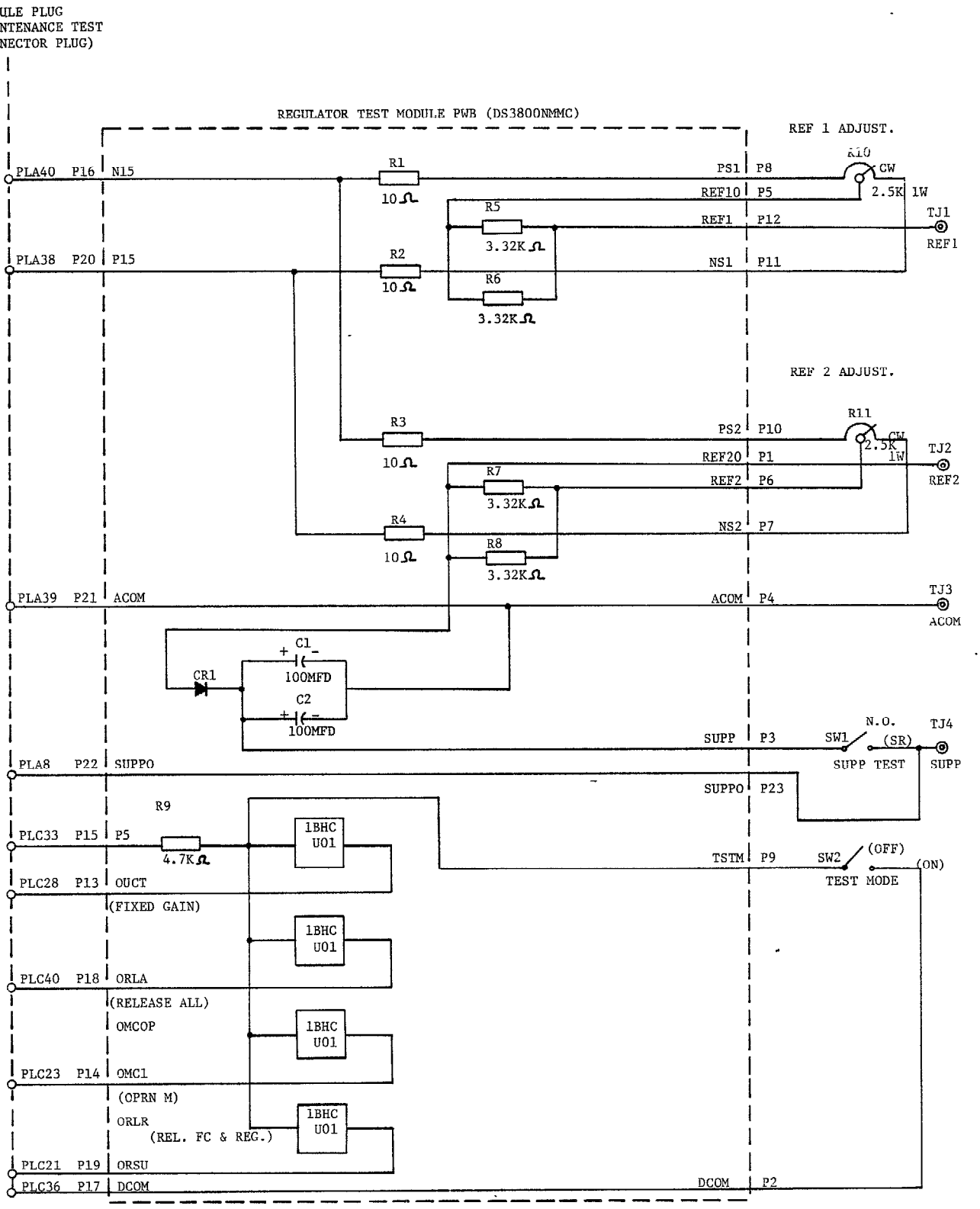


Figure 4. Regulator Test Module Elementary Diagram

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k. RTM CONNECTED TO SUPPRESSION CIRCUIT

If the RTM is wired to the suppression circuit via the card rack backplane, the operation of the RTM test circuit is determined by the RTM's ability to generate a suppression condition at the firing circuit card. To test the RTM, proceed as follows:

(1) Rotate the REF 2 pot fully clockwise (to apply maximum voltage across C1 and C2).

(2) Depress the SUPP switch (This will cause the firing circuit suppression LED to be extinguished).

RTM NOT CONNECTED TO SUPPRESSION CIRCUIT

If the RTM is not connected to the suppression circuit, the checkout necessitates connecting a resistor (100 ohms to 1000 ohms) between the SUPP test jack and ACOM, then observing the voltage at the SUPP test jack. To test the RTM, proceed as follows:

(1) Rotate the REF 2 pot fully clockwise (to apply maximum voltage across C1 and C2).

(2) Depress and hold the SUPP switch closed (down position). The SUPP output should decay to 0.0V (due to the discharging of C1 and C2).

Move the positive lead to the voltmeter from the SUPP test jack back to the REF 2 test jack and observe a voltmeter reading of -14.75V DC (+.75 VDC/-.20 VDC) adjust REF 2 potentiometer for a meter reading of 0.00 volts. Release (open) SUPP switch.

6. OPERATING INSTRUCTIONS.

A. General Purpose References. The purpose of the REF 1 and REF 2 outputs is to provide voltage reference sources for various test points throughout the drive electronics. These reference voltages are developed between the REF 1/REF 2 test jacks and the ACOM jack. These voltages are adjustable by the REF 1/REF 2 potentiometers on the RTM front panel. The use of these test references requires jumpering between the RTM test jacks and various test points on the drive electronics card under test. The REF 1 and REF 2 outputs provide maximum voltages of $\pm$14.95 VDC and a maximum current of 9.0 ma at $\pm$14.95 VDC.

B. Suppression Test References. The purpose of the suppression test reference output is to provide a high current - short duration pulse to test the firing circuit suppression circuit. The voltage pulse amplitude is adjustable by the REF 2 potentiometer. No external connections of the RTM are required to charge the suppression test capacitors.

There are two ways of delivering the voltage pulse from the RTM to the firing circuit suppression circuit as follows:

(1) Jumpering between the SUPP test jack and the positive side of the burden resistor on the high voltage board.

(2) By having an internal connection to the firing circuit card via the card rack back-plane wiring.

C. Test Mode. The purpose of the test mode is to allow testing of the drive electronics without operating the motor (motor contactor open and without feedback signals to the regulator).

D. Suppression Test. The suppression circuit test in the drive electronics is accomplished by verifying that the test pulse generated by the regulator test module will trip the firing circuit suppression circuit. When the suppression circuit is tripped, the suppression LED on the firing circuit card will be extinguished. A toggle switch on the firing circuit card is provided to reset the suppression circuit which will illuminate the LED indicator. To check the module suppression circuit, proceed as follows:

(1) Set the REF 2 output to its maximum positive value, which will supply approximately +11.74 VDC to SUPP circuit.

D. Continued

(2) Reset the firing circuit suppression circuit if it has been tripped.

(3) Initiate the suppression test by depressing the suppression toggle switch on the module. The firing circuit should suppress.

(4) In order to determine the firing circuit suppression level, it is necessary to repeatedly adjust the REF 2 setting above and below the suppression trip level threshold. A high REF 2 setting which extinguishes the suppression LED indicator and a lower REF 2 setting which does not extinguish the suppression LED indicator should be determined by the repetitive adjustments described above. The suppression level will be in error by a diode drop, approximately 0.5 volts less the value set at the REF 2 jack.

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